

Moritz Schwerer

Robot Learning Research Engineer | RL Simulation, Human Motion Data, Real-Robot Evaluation

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Profile

I build robot learning systems end to end: PyTorch training code, simulated environments, robot assets, data and observation design, experiment tooling, failure analysis, and real-hardware validation. My work focuses on RL for embodied agents, imitation learning from human motion data, sim-to-real transfer, scalable evaluation loops, and reliable robot behavior under perception, actuator, and timing constraints.

Experience

Master's Thesis, Hochschule Offenburg | End-to-End RL for Autonomous Humanoid Football

- Implemented the full RL pipeline using PyTorch and Isaac Gym: recurrent PPO-style training, asymmetric actor-critic learning, rollout/storage code, reward design, experiment configuration, evaluation, and deployment tooling.
- Constructed simulation tasks for a Booster T1 humanoid policy that maps robot state, ball perception, and task observations to behavior: search, approach, alignment, kicking, reset, and continued play without external motion capture or manually triggered subskills.
- Designed policy inputs, objectives, and training assumptions for partial observability and real-world uncertainty: deployment-oriented actor observations, privileged critic inputs, delayed/noisy ball observations, sparse goal rewards, auxiliary latent prediction, and action distillation.
- Integrated learned control with real-robot perception and runtime constraints from a head-mounted RealSense camera, ball detection, odometry, compact task observations, actuator behavior, perception delay, and ball-dynamics delay.
- Built simulation and hardware evaluation loops for first-goal reliability, repeated-reset play, alignment quality, kick direction, long-horizon degradation, and controlled-position scoring; validated 26/30 hardware goals.

Student Research Assistant, Hochschule Offenburg | Humanoid Robot Learning

- Worked across the robot learning stack for Booster T1 and Unitree G1: robot assets, collision models, normalized observation/action interfaces, domain randomization, system identification, evaluation scripts, and deployment.
- Developed simulation-trained locomotion controllers that transfer across different robot morphologies, actuator behavior, action spaces, and deployment stacks.
- Built imitation-learning infrastructure for whole-body humanoid skills using consumer motion capture, AMASS / SMPL-H human motion data, ReActor-style retargeting, mocap calibration, Isaac Lab environments, and custom RL algorithms.
- Created deployment-oriented policy evaluation tooling for sim-to-real transfer, robot-specific constraints, actuator/latency effects, and real-world behavior debugging.
- Contributed to a humanoid football system that placed 2nd in the 3v3 football competition at the World Humanoid Robot Games 2025 in Beijing using onboard sensing and real-time team behavior.

Research Alignment

Robotic generalists, multimodal robot learning, vision-language-action policies, RL post-training, scalable autonomy, simulation for robot learning, human data for robot use cases, foundation models for physical AI, and evaluation of long-horizon autonomous behavior.

Technical Skills

Robot learning: RL policies, imitation learning, recurrent actor-critic methods, asymmetric critics, auxiliary losses, action distillation, reward design, behavioral evaluation.

Simulation and data: Isaac Gym, Isaac Lab, MuJoCo, MJCF/URDF assets, AMASS / SMPL-H human motion data, motion retargeting, domain randomization.

Robotics systems: perception-conditioned control, normalized observation/action interfaces, low-level PD target control, system identification, actuator and latency modeling, deployment debugging.

Software: Python, C++, PyTorch, ROS/ROS2 workflows, FastAPI, SQLite, Git, Linux.

Education

M.Sc. Computer Science, Hochschule Offenburg, Germany

Thesis: "Learning Robust Goal-Directed Kicking for Real-World Humanoid Robots via End-to-End Reinforcement Learning"